25901929 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/MGU-BBA (Honours)/MGU-BCA (Honours) Special Examinations, July 2026 (for 2024 Admission Regular Examinations, October 2025)

THIRD SEMESTER

Value Addition Course (VAC) - MG3VACMAT200 - MASTERING PROBLEM SOLVING THROUGH VEDIC MATHEMATICS

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 15 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Questions

Each question must be solved using the relevant Vedic Sutras with proper steps

Answer any **5** questions

Each question carries **2** marks

1. Using Ekadhikena Purvena method, find the sum of 38,93,75,98 [S] / [CO2]
2. Subtract 138 from 224 using Digit Separator Method. [U] / [CO1]
3. Using Nikhilam Navatascaramam method, subtract 3187 from 9263 [S] / [CO2]
4. Define Ekadhikena Purvena in multiplication. [U] / [CO1]
5. Estimate the Ekadhika of 37. [A] / [CO5]
6. Express 250 as a percentage of 50. [U] / [CO8]
7. Find $(x^2 - 2x - 3) \times (x^2 + x - 1)$ using Urdhva Tiryagbhyam. [A] / [CO7]
8. Find $(2a + 4c)(a + c)$ using Urdhva Tiryagbhyam method. [U] / [CO7]

[2x5 = 10]

Part B

Short Essay Questions

Each question must be solved using the relevant Vedic Sutras with proper steps

Answer any **4** questions

Each question carries **5** marks

9. Multiply 1069 by 111. [S] / [CO2]

10. Multiply 36 by 34. [S] / [CO2]

11. If 192 mangoes can be bought for 15, how many can be bought for 5 [A] / [CO5]

12. A sells a bicycle to B at a profit of 30% and B sells it to C at a loss of 20%. If C pays ₹ 520 for it, at what price did A buy? [A] / [CO5]

13. Find the value of x in $\frac{x-2}{x-3} + \frac{x-3}{x-4} = \frac{x-1}{x-2} + \frac{x-4}{x-5}$ using Sunyam Samyasamuccaye. [U] / [CO1]

14. Solve $8x + 3y = 6$ using *Sunyam Anyat.* [U] / [CO7]
$5x + 6y = 12$

[5x4 = 20]

Part C

Essay Questions

Each question must be solved using the relevant Vedic Sutras with proper steps

Answer any **2** questions

Each question carries **10** marks

15. (a) Divide 10312 by 87. [S] / [CO2]
(b) Divide 38982 by 73.

16. A took a certain sum as loan from bank at a rate of 8% simple interest per annum. A lends the same amount to B at 12% simple interest per annum. If at the end of five years, A made a profit of Rs. 800 from the deal, what was the original sum? [A] / [CO3]

17. Find the cube of 998 and 1003 with base 100 and base1000 [S] / [CO2]

18. Solve the following [A] / [CO3]
$(a)(x+2)(x+8) = (x+3)(x+6)$
$(b)\frac{5x+7}{2x+3} = \frac{4}{3}$

[10x2 = 20]